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A Novel End-to-End Corporate Credit Rating Model Based on Self-Attention Mechanism

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ABSTRACT In traditional financial analysis discriminant analyses with statistic methods are considered as primary and widely applied corporate credit rating bases in which human impact plays an essential role. Machine learning approaches enormously assist credit rating in reducing human interference and improving rating efficiency. However, existing methods did not efficiently leverage the time series features of financial data that is beneficial for the classification tasks. We propose a novel end-to-end architecture, the SMAGRU, based on self multi-head attention with gated recurrent neural networks, which is capable of evaluating a corporate credit rating by capturing the time series features assigned the weights of market benchmarks. Experiments compared with statistic and machine learning baseline models on different datasets show that our model contains superiority in multi-class credit rating classification. The result of ablation experiments suggests that time series features are crucial in classification performance, and self multi-head attention can significantly enhance these features and thereby improve the accuracy and convergence speed. We show that the SMAGRU generalizes well, even on sparse data during K-fold cross validation experiments.

INDEX TERMS Corporate credit rating, multi-class classification, self multi-head attention, GRU, time series.

I. INTRODUCTION

A corporate credit rating is an opinion regarding the assessment of a company's ability to meet its debt obligations according to an independent credit rating agency, such as Moody's, Standard & Poor's (S&P's), and Fitch Ratings. Corporate credit ratings are crucial not only since they are general requirements of public bond issuance and specific loan structures, but that they are considered as efficiently practical guidelines of investment meanwhile provide access to a broader range of lenders and debt products [1]. For instance, despite slight difference between rating agencies, AAA denotes the highest credit quality, C or D often represents the lowest quality or default risk, which describes the range of a regular credit rating scale. Credit rating activities adopting traditional methods are predominantly regarded as financial or mathematical evaluation problems, diversely, credit rating performs as an instance of multi-class classification problems in machine learning.

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The credit rating method undergoing reforms has made significant progress and achieved remarkable results. Analyses methods based on characteristics, financial ratios, factors [1]–[3] and evaluation models like ZETA, KMV, CreditMetrics, and *et al.* [4], [5] have been the fundamental methods in risk prediction also in credit evaluation. Most models are based on data analysis and statistical methods, which significantly improve the accuracy of credit assessment and reduce the subjective factors of prior approaches.

Despite the dominant role in the rating market, these traditional methods have been suffering from the influence of human impacts on credit rating or risk evaluation. Furthermore, great rating agencies tend to emphasize the significance of analysts' subjective judgment in determining parameters or directly credit ratings. Besides, parameter analysis methods like logit regression or KMV model assume the normal distribution of the data to be analyzed, which is not necessarily consistent with real-world data. Moreover, all but a few models in traditional ways excel at tackling linear relationships instead of non-linear ones.

In recent years, with the widespread application of machine learning, many methods in this field have also been

introduced into the research of credit ratings, such as Support Vector Machines (SVM), Decision Tree (DT) and *et al.* Due to the characteristics of machine learning, the introduction of these approaches have greatly improved the drawbacks of previous rating works that relies too much on the manual decision. More importantly, machine learning methods provide high efficiency and accuracy in rating classifications, especially in binary classification tasks with small sample data. However, despite the advantages mentioned above, time serial problems are considered complex works for these methods, and that they do not obtain satisfied enough performance in tackling high-dimensional data and multi-class classifications.

Neural networks (NNs) and deep learning methods, such as multilayer perceptron (MLP) and backpropagation neural network (BPNN), can efficiently process high-dimensional data and non-linear relationships. Nonetheless, BPNN models regularly suffer from slow convergence speed and vibration during training. Both MLP and BP neural networks obtain significantly lower accuracy in multi-class classification than binary classification.

In this study, inspired by transformer [6], a novel end-to-end approach based on multi-head self-attention mechanism and recurrent neural network with gated recurrent units (GRUs), namely self multi-head attentive gated recurrent network (SMAGRU), is introduced in corporate credit rating. This study notably differs from traditional credit rating methods in end-to-end mechanism which cannot be influenced by artificial adjustment [7]. In contrast to priorly evaluating credit rating based on specified period, according to our assumption that a corporate credit rating decision should base on its long-term operating status, this model effectively utilizes the temporal attributes of financial data to classify the credit ratings due to the sequential memory ability of RNN. Moreover, our model can efficiently extract financial indicator features from high-dimensional data and non-linear relationships through self-attention mechanism, which helps model improve rating performance.

Our main contributions are summarized as follows:

- (1) It is the first study to implement multi-head self-attention mechanism in financial analysis, specifically in credit rating, and to demonstrate that self multi-head attention mechanism with outstanding ability in extracting financial element representations enhances the classification performance of the gated recurrent networks in our architecture.
- (2) We treat corporate credit rating under the insight of dynamic financial analysis [8]. The past financial indicators, along with the present ones, help grasp the dynamic financial features [9], and further evaluate a corporate credit rating. Experiments show that leveraging time series features the model can effectively improve corporate rating classification.
- (3) Benefit from the high performance of neural networks in processing high-dimension data and non-linear relations, we proposed an end-to-end architecture which

outputs the rating results directly taking disclosed financial statement values. The architecture is easy to follow and replicate.

- (4) We take financial data of all entities listed in rating agencies to construct each financial indicator vector to simulate the financial status of the entire market in order to generate the more representative indicators the model needs and further generate the weights of each indicator replacing the man-set weights in prior works.

II. RELATED WORKS

A. CORPORATE CREDIT RATING

In commercial applications and economic studies, credit rating is mainly determined by subjective judgment and statistical-based models. Credit appraisal or rating initiate with basic qualitative and quantitative assessment measurement, for instance, 5C's, 5P's, LAPP, financial ratio analysis, factor analysis [1]–[3], [10], [11]. The drawback of this bunch of methods is that the dominant subjective factor and simple collocation of discriminating factors cause a tremendous difference and low efficiency in evaluation.

The credit evaluation models based on statistics have primarily solved the problems of human factors and work efficiency. Multiple discriminant analysis (MDA) [12], Logit regression [13], Probit model [14], [15] have performed good accuracy and robustness in predicting bankruptcy of enterprises or banks. However, they have high requirements for the distribution of sample data and cannot manage high-dimensional, complex data. On the basis of univariate analysis [1], leveraging multiple income and balance sheet values to measure corporate financial health, Altman developed his multivariate discriminant model - Z scores [10] into ZETA model [4], where a Z-Score is a statistical measurement describing a value's relationship to the mean of a group of values. Despite full application in risk analysis and credit rating so far, its major drawback lies in dealing with non-linear relations extensively existing in credit rating processing. Projection pursuit regression [16] projects non-linear as linear problems, however, containing limited ability in solving non-linear problems.

In contrast with statistical techniques, AI techniques do not acquire a certain assumption of data distributions. Furthermore, AI architectures outperformed statistical ones in dealing with non-linear pattern classification missions, especially with high-dimensional data, such as credit scoring problems [17]. Odom & Sharda first introduce a neural network for corporate bankruptcy prediction [18]. Afterward, NNs are implemented in Bankruptcy forecast [19], Business failure prediction, Debt risk assessment, Security Market Applications, Financial forecasting, and *et al.* [20], [21]. However, NNs, namely MLP, BPNN, have been criticized for their difficulty in deciding learning rates, ease of being stuck on local minimums, overfit problems, and time-consuming [22]. The traditional BPNN algorithm demands a large number of parameters in network training, and it

is easy to be trapped into local optima [23]. Cluster analysis [24]–[27], K Nearest Neighbor (KNN) [21], [28]–[31], Support vector machines (SVMs) [32], [33] have been extensively applied for credit ratings or scoring, while previous comparison studies have showed that SVMs outperform NNs on credit ratings [17], [34]. Decision Tree (DT) is another widely used technique in this field due to its white-box feature and fast construction [35], [36]. Despite the improvement from ID3 to GDBT, DT models are not good at processing high-dimensional, sparse data, and data with strong feature correlation [37].

B. ATTENTION MECHANISM

Attention based models are recently widely used in machine translation, image caption, entailment reasoning, speech recognition, abstract generation. The attention mechanism was proposed to reduce the computational cost by applying convolutional neural networks to deal with static image classification and dynamic visual control problems. Via the attention mechanism, the model selects a single different location within images or video frames according to past information or demands of the task, instead of processing a whole picture at once [38].

The very core of attention model applying in encoder-decoder architecture is align function or score function, which is designed to calculate the similarity or relevance of two vectors according to the demands, more often the similarity between input word and predicting output word in translating mission. The procedure is summarized as below:

$$\text{Attention} = \text{Similarity/Relevance}(\text{Source}, \text{Target}) \quad (1)$$

Bahdana *et al.* [39] implemented in machine translation model, a multilayer perceptron, as an align function, which is called additive attention or concatenation [40]. While Luong *et al.* introduced three content-based alternatives of align function, comparing the current target hidden state with each source hidden state via dot, general, concatenation, and another location-based function, which computes the sole align score of the hidden state. [40]

The model attending to all the source hidden states is regarded as Global Attention, while partial the source hidden state - Local Attention. Both Global Attention by Luong and Additive Attention by Bahdana are all classified as soft attention.

Following the Bahdana Attention model, Yang *et al.* (2016) proposed Hierarchical Attention Network (HAN) to construct a document representation by capturing word attention and sentence attention. With the similar spirit of hierarchical attention, some studies were introduced in grammatical error correction [41], sequence to sequence learning [42], image recommendation [43].

We can also see self-attention models implement in many fields besides machine translation. The attending process,

$$a_{ij} = \text{softmax}(f(u, v_i)) \quad (2)$$

where v_i is matched with u contained within the sequence of v_i is regarded as a procedure of self-attention, or intra-attention, otherwise, with u outside the sequence of v_i – extra attention, which is the attention mechanism we often refer to. Function f is variable depending on the problems to be solved.

To construct an encoder-decoder architecture in machine reading, Cheng *et al.* (2016) [44] combined an intra-attention layer on an improved adaptive Long short-term memory (LSTM) structure. Parikh *et al.* (2016) [45] used the similar intra-attention model to compute the relationship between words within a sentence, then to determine the entailment or contradiction relationships between a premise and a hypothesis. Sha *et al.* (2016) [46] implement a similar construction attention model to capture important information on the premise. Lin *et al.* (2017) [47] implemented self-attention model on a bi-LSTM to extract the sentence embedding.

Except for a few cases, most studies containing attention models have adopted recurrent networks. Architectures with recurrent networks are superior in dealing with sequential missions rather than paralleled problems. Aiming to avoid the sequential processing coupled with its complexity of calculation, Vaswani *et al.* (2017) [6] proposed a novel architecture preventing recurrence and allowing sufficiently more parallelization. In the encoder-decoder construction named Transformer. Hereafter, a few alternatives were presented, such as Bidirectional Encoder Representations from Transformers-BERT [48], deep Transformer [49], Transformer-xl [50]. At the same time, some optimized approaches of Transformer were introduced [51], [52], and the same spirit was shared in image processing [53], [54].

C. CLASSIFICATION BASED ON RECURRENT NEURAL NETWORKS

Numerous prior classification tasks are processed by MLP based models. However, recurrent neural network, which excels at mining long-distance discrete information [55], specializes in a continuous-time series problem. With memory ability, it can use all information of context to predict the probability of the next output word [56] and is mainly used for processing text sequence with positional information [57], [58]. However, on the occasion that the input sequence is further expanded, the backpropagation of RNN causes problems with exploding or vanishing gradients. Long short-term memory units are brought out based on RNN to solve this problem [59], [60]. Despite the excellent performance in memorizing, yet LSTM architecture contains the problem of computational complexity. Aiming to simplify the layout of the model and reduce the cost of calculation, while obtaining the same performance [61], a gated recurrent unit (GRU) based on RNN has been proposed [62]. Comparing with classical LSTM network holding three gates and two hidden states, update gate and reset gate and one hidden state coupled in GRU network are introduced as filters deciding to skip or keep information [63].

RNNs along with its alternatives have been commonly used frameworks in machine translation [64], constituency parsing [65], language modeling [66] and recognizing textual entailment [67]. With the advantage in ceaseless sequence, RNNs, especially LSTM or GRU, are outstanding in processing classification tasks related with sequential data, namely, tackling relation classification [55], [68], [69], sentiment classification [70], [71], document classification [72], signal classification [73]–[75], sound events classification [76], long time series land cover classification [77], product quality prediction [78], rating classification [79].

In order to enhance features or capture representations, recently recurrent networks with attention mechanism combined in perform well in text classification [80], sentiment classification [60], [81], [82], document classification [83], malware classification [84], signal classification [54], question classification [85], relation classification [86], [87].

III. METHODOLOGY

In this study, we take financial data of all enterprises from the three achieved rating lists of different credit rating agencies separately to simulate the three market statuses. In contrast with most RNN or CNN models with attention mechanism that share the same inputs with attention module, we differ inputs vectors of weighting module from the rating classifier.

Aiming to extract the feature of each financial statement element of the whole market, the values of all companies under a single financial statement element are pack into a vector. In this study, 20 most representative financial elements are elected, which means 20 vectors per quarter are constructed from which self-attention mechanism capture the financial element features as the weights assigned to classifier. We accessed the financial data within t_n quarters, that is, n columns of self-attention vectors will be computed, which stands for the benchmark weights of the entire market.

During the classification, the time series features are expected to obtain. In this scenario, the values under 20 financial elements of a single company are packed into vectors according to the time steps, one of the quarters. The sequence of vectors which are element-wise producted with the corresponding weight vectors is as input token of the classifier to predicting the proper credit rating. The figure 1 depicts the overview of the SMAGRU architecture.

A. FINANCIAL STATEMENT ELEMENT SELECTION

In traditional statistical, financial analysis financial ratio framework [88] still takes the dominant role in credit assessment. Affected by this, studies on credit rating with machine learning approaches tend to take financial ratios as variables. When dealing with large data scale and overabundant variables, or dimensionality reduction is required due to algorithm restrictions, most studies will adopt various methods, such as factor analysis [11], Analytic Hierarchy Process (AHP) [89], Analysis of Variance [17] and at el., to reduce input variables that are not statistically related to ratings.

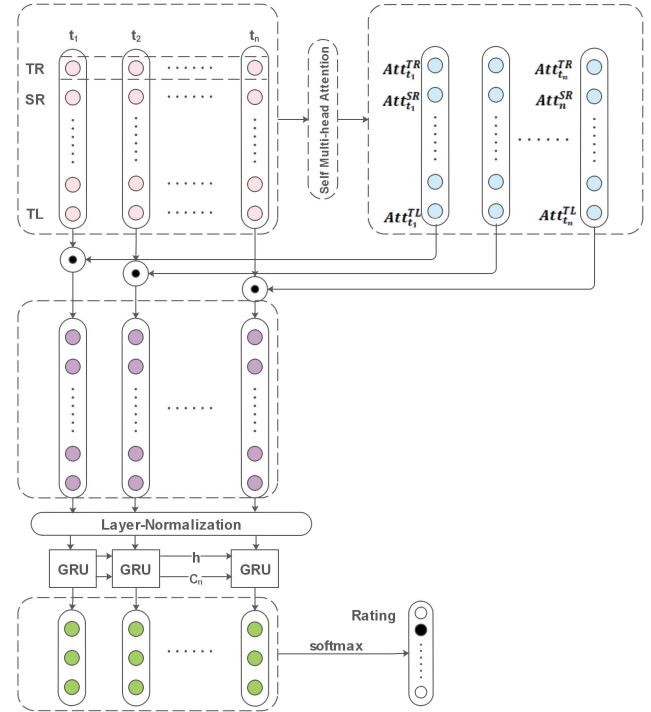


FIGURE 1. The architecture of self multi-head attentive gated recurrent neural network - SMAGRU.

The financial ratio analysis has widely applied to measure the abilities of enterprise operations, such as profitability, liquidity, management efficiency, leverages, and valuation & growth. For instance, in Profitability Ratios, the gross profit rate is calculated by dividing gross profit by net sales, and gross profit is equal to net sales minus cost of sales [90]. Given all the ratios are the result of computations, if specific ratios along with financial element values are put into models as inputs, as shown in some studies [17], it might result in unexpected effects like feature cross in machine learning. Considering the inexplicability of neural networks, we prudently decided to exclude all the financial ratios from calculations as inputs. Furthermore, one of the main purposes of our study is to capture the representation of each financial element, the separate element is required. One more essential motivation doing so is to make data collecting easier, for all financial statement sheets are disclosed.

For reflecting above-mentioned corporate operation abilities, we choose all the necessary financial statement elements used for calculating financial ratios instead of correlation analysis, as follows:

B. FEATURE SCALING

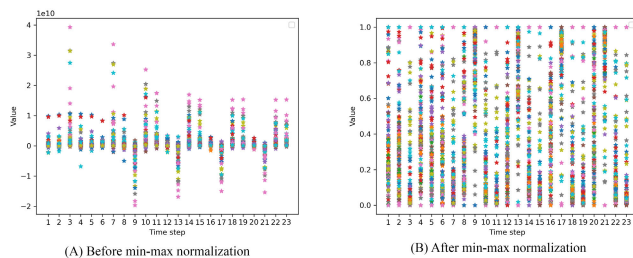
Input values in our model vary greatly. Moreover, containing negative values and zeros. A feature scaling method thus should be implemented before calculations. We apply different data normalization strategies in different modules in our architecture: min-max normalization in attention module, layer normalization in the GRU module.

TABLE 1. Selected financial statement elements as input variables reflecting corporate operating abilities.

No.	Abbr.	Financial statement element
X1	TR	Total Revenues
X2	SR	Sales revenue
X3	CF	Cash paid for Interest
X4	NI	Net Income
X5	CP	Cash paid for Inventory purchases
X6	CW	Cash paid for Wage expenses
X7	CT	Cash paid for Income taxes
X8	CE	Cash paid for General operating and administrative expenses
X9	CC	Cash receipts from customers and other operations
X10	CS	Cash receipts from Sale of investment securities
X11	CB	Cash receipts from Borrowing
X12	CL	Cash paid for Repayment of loans
X13	NC	Net Cash Flow from Investing Activities
X14	NE	Net increase in cash and cash equivalents
X15	DI	Decrease in inventories
X16	IL	Investment loss
X17	CA	Current Assets
X18	TA	Total Assets
X19	TC	Total Current Liabilities
X20	TL	Total Liabilities

We conduct **min-max normalization** on data before the computation of self multi-head attention. We randomly extract values of twenty financial elements of eight entities as a distribution sample. The substantial difference of distributions before and after min-max normalization can be observed in the Fig. 2. The equation of min-max normalization is as follows:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (3)$$

**FIGURE 2.** The left is the distribution of original values of twenty financial elements of randomly selected eight entities. The right one is the distribution after min-max normalization.

Layer normalization is proposed to reduce the training time, besides the function of feature scaling. Furthermore, this method has a better effect on the validation result, especially fitting the recurrent neural networks [91]. Hence, we implement the Layer normalization on the inputs before the activation function in our GRU RNNs.

Equations below illustrate how the Layer normalization work in an RNN.

For the node at RNN time-step t , the summed input a^t is computed from the current input x^t and the previous vector of hidden states h^{t-1} as follows:

$$a^t = W_{hh}h^{t-1} + W_{xh}x^t \quad (4)$$

where W_{hh} is the recurrent hidden to hidden weights and W_{xh} are the bottom-up inputs to hidden weights.

$$h^t = f \left(\frac{g}{\sqrt{(\sigma^t)^2 + \epsilon}} \odot (a^t - \mu^t) + b \right) \quad (5)$$

$$\mu^t = \frac{1}{H} \sum_{i=1}^H a_i^t \quad (6)$$

$$\sigma^t = \sqrt{\frac{1}{H} \sum_{i=1}^H (a_i^t - \mu^t)^2} \quad (7)$$

where H denotes the number of hidden units in a layer, μ^t and σ^t denote mean and variance in time-step t , all the hidden units in a layer share the same μ and σ , $\odot$ is the element-wise multiplication, b and g are defined as the bias and gain parameters of the same dimension as h_t , ϵ is a very small decimal, to prevent division by 0.

C. WEIGHTING MODULE

Based on prior studies on attention mechanisms, the essential abilities of attention models are to gain the similarity or relevance of two references. Many studies have utilized this feature to capture the representations of characters, words, sentence, images *et al.* In our study, we aim to capture the financial representation of a particular financial element among others of the entire market which is leveraged as weight assisting the credit rating classification. Transformer has been proven to comprise the more outstanding feature extraction ability by virtue of self-attention mechanism [92] and to remarkably allow more parallelization and consideration of relevance between target and all inputs [6]. We thus establish the weighting module based on self multi-head attention mechanism as shown in the Fig. 3.

Given input F^n ($n = \text{numbers of financial elements}$, $n \leq 20$, $n \in \mathbb{N}^*$). F^n is vector with a dimensions which is constructed by certain financial statement element values of the entire market in D1 dataset, while b dimensions in D2 dataset, and c dimensions in D3 dataset ($a = 2619 = \text{total enterprise number of D1}$, $b = 910 = \text{total of D2}$, $c = 515 = \text{total of D3}$).

In this section, we aim to calculate the feature Att of F^n , that is, the weight of a particular financial statement element in the whole market. Inspired by Vaswani *et al.* [6], we hence build a model consisting of a stack of 6 duplicate layers consisting two sub-layers: multi-head self-attention mechanism and fully connected feed-forward network. By contrary with the vanilla transformer, given the sequential features are not required, the position encoding is not included in our model.

Each input projects three its representations called query, key, and value. An align score e_{qk} is the result of scaled

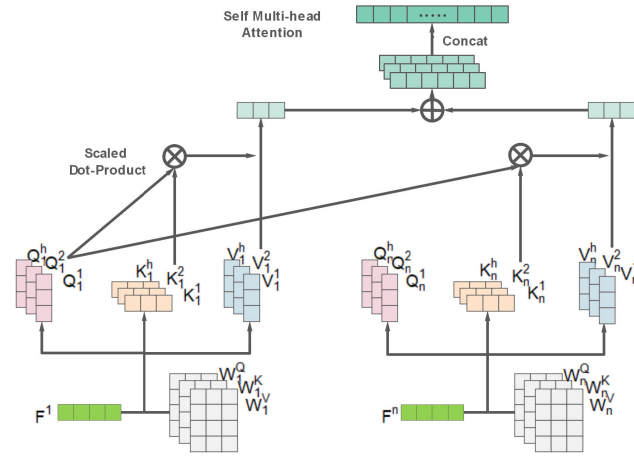


FIGURE 3. The architecture of self multi-head attention mechanism.

dot product by $\frac{1}{\sqrt{d_k}}$ between a query and all keys. The softmaxed align score is multiplied as weight with the corresponding value, the self-attention SA is the element-wise sum of weighted values. In order to capture deeper features of financial statement element, the multi-head attention Att generated via weighting self-attention is taken as the module output, namely, the financial element weight. The queries, keys and values are packed into sets of matrices Q, K, V, and computed as follows:

$$e_{qk} = f(Q, K) = \frac{QK^T}{\sqrt{d_k}} \quad (8)$$

$$SA(Q, K, V) = \sum_n SoftMax(e_{qk}) V \quad (9)$$

$$Att = Concat(head_1, \dots, head_h) w^O \quad (10)$$

where $head_i = Attention(QW_h^Q, KW_h^K, VW_h^V)$, $W_h^Q \in \mathbb{R}^{d_{model} \times d_k}$, $W_h^K \in \mathbb{R}^{d_{model} \times d_k}$, $W_h^V \in \mathbb{R}^{d_{model} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{model}}$. In this work, we employ $h = 8$ parallel attention layers or heads. For each of these, we use $d_k = d_v = d_{model}/h$.

We repeat the procedure mentioned above to capture the weight of each financial element between other elements in different time steps. The results are as below:

$$\begin{bmatrix} Att_{t_1}^{TR} & \dots & Att_{t_n}^{TR} \\ \vdots & \ddots & \vdots \\ Att_{t_1}^{TL} & \dots & Att_{t_n}^{TL} \end{bmatrix} \quad (11)$$

The weights in different time steps are capable of representing the benchmark of the corresponding financial element of the entire market, thereby are assigned to regulate corresponding values of financial statement elements of each company prepared to input to the classifier.

D. DYNAMIC CREDIT RATING CLASSIFIER

In traditional Dynamic financial analysis, the industry average ratio is usually used as a financial evaluation benchmark

to compare the ratio of a specific company at a specific point in time [7]. Excessive deviation of a company's financial ratio from the industry average is considered undesirable [9], and it can be found that a large number of entities tend to smooth the financial ratio and further to approach the industry average within a required period [7]. The purpose of the financial ratio adjustment method is to identify and eliminate human manipulation. The usual static comparative analysis will focus on a specific time point. Instead, dynamic financial analysis advocates the regression of current ratios and past ratios, which helps contain the dynamic nature of the company's financial ratios and the adjustment process. In our study, we argue that the assessment of corporate credit rating should base on analysis of dynamic long-term financial performance, which is proper to be considered as a time series problem. As mentioned in related works, LSTM and GRU RNNs contain notable memory capacity in handling time series missions, thus being the optimal basis of the classifier module.

RNN with LSTM cells introduces a gating mechanism (input gate, forget gate, output gate) to alleviate the long-term dependence problem in recurrent neural networks. Eqs. 20-22 summarize how the LSTM cell computes long-term, short-term states, and its output at each time step for a single instance.

$$\begin{bmatrix} i_t \\ f_t \\ o_t \\ \hat{c}_t \end{bmatrix} = \begin{bmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{bmatrix} W \cdot [h_{t-1}, x_t] \quad (12)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \hat{c}_t \quad (13)$$

$$h_t = o_t \odot \tanh(c_t) \quad (14)$$

By contrary with RNNs, in a basic LSTM cell, the hidden state is split into two vectors: h_t and c_t , which respectively represent the short-term state and the long-term state. The current input vector x_t and the previous short-term state h_{t-1} are fed to four different fully connected layers. One layer outputs $\hat{c}_t$ analyzing the current inputs x_t and the previous (short-term) state h_{t-1} . The other three are gate controllers. The forget gate controlled by f_t controls which parts of the long-term state should be erased. The input gate controlled by i_t controls which parts of $\hat{c}_t$ should be added to the long-term state. Finally, the output gate controlled by o_t controls which parts of the long-term state should be read and output at this time step (both to h_t) and y_t .

The GRU cell simplifies the structure of the LSTM cell and performs as well as the latter in most cases. In a GRU cell as shown in the Fig. 4, the short-term state and the long-term state merge into a single vector h_t . Output gate no longer exists, instead, the state vector h_t is output at each time step

In time step t , x_t is plugged into the network unit, while multiplied by the weight W_z . The same goes for h_{t-1} which holds the information for the previous time step $t - 1$ and is multiplied by the weight U_z . Both results are added together, and a sigmoid activation function is applied to squash the

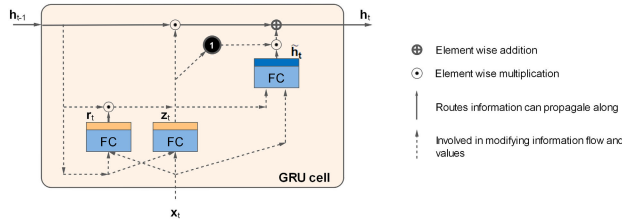


FIGURE 4. The architecture of gated recurrent neural network – GRU RNN.

result between 0 and 1. The gained above z_t is update gate which decides how much of the past information needs to maintain and be passed along to the future.

$$z_t = \sigma(W_z x_t + U_z h_{t-1}) \quad (15)$$

The reset gate r_t helps the model to determine how much the past information to forget.

$$r_t = \sigma(W_r x_t + U_r h_{t-1}) \quad (16)$$

A new current memory $\tilde{h}_t$ is calculated as follows by utilizing the reset gate to keep the relevant information from the past. The element-wise product between the reset gate r_t and $U h_{t-1}$ determines what to remove from the previous time steps.

$$\tilde{h}_t = \tanh(W x_t + r_t \odot U h_{t-1}) \quad (17)$$

The final memory h_t of t time step is combined from the current memory $\tilde{h}_t$ and previous memories h_{t-1} . The update gate is implemented to choose information from $\tilde{h}_t$ and h_{t-1} .

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \quad (18)$$

In this case, if the model learn to set the update gate, the vector z_t , close to 1, which means keeping a majority of the previous information, $(1 - z_t)$ will be close to 0, which means the model will ignore a large percentage of the current content.

Considering the sequential feature of GRU RNN, we organize each financial statement element values in chronological order: $(TR_{t_1}, \dots, TR_{t_n})$. The chronological values of certain element dot-multiplied by corresponding weights gained from dynamic weighting module are fed into classifier as inputs as follows: $(Att_{t_1}^{TR} \odot TR_{t_1}, \dots, Att_{t_n}^{TR} \odot TR_{t_n})$.

E. MODEL TRAINING

We append a softmax layer at the end of the model to convert the output vector y to the probability distribution of the prediction.

$$\text{SoftMax}(y)_i = y'_i = \frac{e^{y_i}}{\sum_{j=1}^N e^{y_j}}, \quad i = 1, 2, 3 \dots N. \quad (19)$$

On this basis, the distance between the probability distribution of predictions and the ground truth which can be calculated by cross-entropy.

In this model, we apply categorical-cross-entropy, which is considered more efficient in multi-class classification tasks in Keras as a loss function.

$$L_i = - \sum_j t_{ij} \log(p_{ij}) \quad (20)$$

where $p_{ij} = f(y) = y'_i$ = Probability of i_{th} element at grade j , t is one-hot encoded the ground truth vector, i denotes the data point, and j denotes the grade. In the experiment, the stochastic gradient descent algorithm is used to minimize the loss function.

IV. DATASETS

A. CORPORATE CREDIT RATING

The credit rating symbols used by domestic rating agencies are mainly formulated following the unified requirements (as shown in the Tab. 2) of the “Specification for Credit Rating in the Credit Market and Interbank Bond Market, Part 2: Specification for Credit Rating Business” issued by the People’s Bank of China in 2010 [93]. The credit evaluation system of Jian Rating (Jian FinTech Corporation) adopted in this study (as shown in Tab. 2) is formulated according to this standard. There are mainly three rating agencies that enjoy an authoritative status in the industry: S&P’s, Moody’s, and Fitch. The three agencies obtain slight differences in credit rating symbols and corresponding worthiness. The issuer credit ratings the three agencies issued are equivalent to corporate credit ratings mentioned in this study, as shown in the Tab. 3.

B. DATASETS

According to the credit rating reports in 2020 of different rating agencies, we construct three datasets with values from disclosed corporate financial statements with credit rating labels. Labels of the D1 dataset are from corporate ratings of Chinese rating agency Jian rating, which evaluates all Chinese A-share companies. The rating is rich in grades ranging from AAA to ST. The risk distribution is relatively uniform and covers more company entities, which helps to capture the characteristics of enterprises with a higher risk of default. However, more classes mean a greater test of the classification performance of the model. As shown in Tab. 4, the D1 dataset includes 1,204,740 values of 2619 companies in 10 ratings of 23 quarterly financial reports from March 2014 to December 2019.

The credit rating labels of the D2 dataset are obtained from the bond issuer ratings of the Shanghai Stock Exchange, Shenzhen Stock Exchanges, and ChinaBond.com.cn in 2020, which are the three public bond trading markets in China. All the credit ratings are issued by the three largest credit rating agencies in China, China Chengxin International Credit Rating, Dagong Global Credit Rating Group, and United Credit Ratings. As the “Shanghai Stock Exchange Corporate Bond Listing Rules (2015 Revision)” requires that issue rating of AAA is the condition for the public listing of bonds, while the listing requirements of “Shenzhen Stock Exchange Corporate Bond Listing Rules (2018 Revision)” are both

TABLE 2. Credit ratings and creditworthiness of the people's bank of China and Ji'an rating.

People's Bank of China	Jian Rating	Credit worthiness
AAA	AAA	the highest debt repayment ability, positive operating recurring state.
AA	AA	very good debt repayment ability, positive operating recurring state.
A	A	good debt repayment ability, positive operating recurring state.
BBB	BBB	average debt repayment ability, positive operating recurring state.
BB	BB	low debt repayment ability, poor operating recurring state, a certain degree of risk.
B	B	poor debt repayment ability, troubled operating recurring state, high risk.
CCC	CCC	very poor debt repayment ability, troubled operating recurring state, very high risk.
CC	CC	severe inability to repay debt, troubled operating recurring state, extremely high risk.
C	C	extremely poor debt repayment ability, negative operating recurring state, on the verge of bankruptcy.
	D	specially refers enterprises in stock market, on the verge of ST.
	ST	operating loss for two consecutive years, warning of exiting the stock market.

TABLE 3. Credit ratings and creditworthiness of S&P's, Moody's and Fitch.

MOOSY'S	STANDARD & POOR'S	Fitch Ratings	Credit worthiness
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC+	Substantial risks
Caa2	CCC	CCC	
Caa3	CCC-	CCC-	
Ca	CC	CC	Extremely speculative
	C	C	Default imminent
C	RD	RD	In default
/	SD	D	
/	D	/	

issue ratings and issuer ratings reach AA, so the rating distribution of D2 dataset (as shown in Tab. 4) is more concentrated, lack of some ratings. D2 dataset contains 418,600 values of 910 companies in 8 ratings 23 quarterly financial reports from March 2014 to December 2019.

With the same method, we use the values of American enterprises to construct the D3 dataset (as shown in Tab. 4) in which credit rating labels from S&P's and Moody's. D3 dataset contains 51,500 values of 515 companies in

10 ratings 5 quarterly financial reports from March 2019 to March 2020.

V. EXPERIMENTS AND DISCUSS

A. EXPERIMENTAL SETUP

We perform experiments to compare the classification performance of our model with other baseline models on three datasets. Accuracy measures are utilized to score

TABLE 4. Corporate credit rating distribution in datasets.

D1		D2		D3	
Rating	Number of companies	Rating	Number of companies	Rating	Number of companies
AA	55	AAA	129	AAA	3
A	250	AA	519	AA	15
BBB	247	A	215	A	74
BB	529	BBB	29	BBB	189
B	519	BB	4	BB	140
CCC	293	B	1	B	75
CC	387	CC	2	CCC	14
C	172	D	11	CC	2
D	34			C	2
ST	133			D	1
SUM	2619		910		515

TABLE 5. Credit rating classification accuracies and AUC values of our model and baseline models.

Models	Accuracy			AUC		
	D1	D2	D3	D1	D2	D3
LR	0.5493	0.5379	0.3413	0.550±0.001	0.536±0.002	0.338±0.000
KNN	0.8080	0.7856	0.6529	0.813±0.003	0.798±0.002	0.332±0.018
SVM	0.7648	0.7199	0.3541	0.782±0.002	0.743±0.003	0.354±0.005
DT	0.8349	0.8351	0.6807	0.185±0.003	0.179±0.009	0.681±0.002
GBDT	0.8001	0.7603	0.6243	0.762±0.002	0.758±0.004	0.620±0.012
BPNN	0.2035	0.1965	0.3609	0.680±0.024	0.750±0.016	0.560±0.029
SMAGRU	0.9333	0.9370	0.9068	0.93±0.014	0.88±0.018	0.75±0.025

the performance between SMAGRU and baseline models. To evaluate the time series characteristic and the effect of attention, we then run ablation experiments within Unidirectional GRU without Self multi-head attention (GRU), Self multi-head Bidirectional GRU (SMABiGRU) and Self multi-head Unidirectional GRU (SMAGRU). In order to further verify the generalization ability of SMAGRU, we perform k-fold validation experiments.

In the credit rating classifier module, the mini-batch method is applied for iterative training, and each batch size is 23. We implement the Stochastic Gradient Descent (SGD) algorithm to minimize the loss function, Adaptive moment estimation (Adam) [94] algorithm to adaptively update the parameters and the learning rate is 0.01, dropout method [95] to prevent the overfitting and dropout rate is 0.2. In the models training phase, we have assessed different hyper-parameters of epochs, layers, and cells. The follow settings have achieved the best accuracy: epoch = 100, GRU layer = 4, cell = 200.

B. COMPARISON BETWEEN OUR MODELS AND BASELINE MODELS

In order to evaluate the performance of the model, our model is compared with the following baseline models on both D1 and D2 datasets. Respectively, the classification performances and AUC values of the models are given in Tab. 5.

LR with L2 regularization is the default setting in toolkit scikit-learn 0.22.2 in Python 3.7. In order to be capable of classification tasks, LR is set to be with L1 regularization.

KNN adopts the default setting in toolkit scikit-learn.

SVM is set with strategy One-Vs-The-Rest (OVR), which is most commonly used for multi-class classification and is the default strategy of scikit-learn.

DT's version in scikit-learn is ID 3.0.

GBDT is Gradient Boosting Decision Tree. Given ID 3.0 is an early version, GBDT is added into baseline models.

BPNN adopts the same hyperparameters with GRU, SMABiGRU, SMAGRU.

SMAGRU is Self multi-head + Unidirectional GRU.

We can see from Tab. 5 that SMAGRU contains the best performance in classification and the highest AUC value. Among baseline models, nonparametric algorithms, such as KNN, DT, and GBDT, gain an outstanding classification result. However, among these three models, it can be observed that on the D3 dataset, which consists of the least, sparse, feature correlated data classification results turn out lower than on D1 and D2, which is consistent with the results of the paper [37]. As an improved version, GBDT obtains the lower accuracies than DT on all three datasets. Notably, the AUC value of DT is substantially lower than all models, including GBDT, which means the DT model is not robust in

multi-class classification. In our classification experiments, the classification performance of BPNN without optimization is enormously lower than other models, which demonstrate insights of prior works [17], [23], [34].

C. ABLATION EXPERIMENTS

To evaluate the effects of the self multi-head attention module and time series feature, we present an ablation test on the proposed model in dataset D1. Tab. 6 shows the rating classification results of unidirectional GRU without Self multi-head attention (GRU), self multi-head attentive bidirectional GRU (SMABiGRU), and self multi-head attentive unidirectional GRU (SMAGRU). From the results, it can be seen that classification accuracies of models with attention function, SMABiGRU and SMAGRU, are significantly higher than that of GRU without attention. At the same time, it can be noticed that SMABiGRU performs weaker than SMAGRU, especially on the D3 dataset.

TABLE 6. Comparison of effects of self multi-head attention and bidirectional modules.

Model	Modules	D1	D2	D3
GRU	Ø-uni-GRU	0.6042	0.4256	0.6334
SMABiGRU	SMA-bi-GRU	0.8298	0.8370	0.6635
SMAGRU	SMA-uni-GRU	0.9333	0.9370	0.9068

1) EFFECTIVENESS OF ATTENTION MECHANISM

According to the comparison results between SMAGRU and GRU without attention, the self multi-head attention mechanism has achieved two improvements.

As shown in (as shown in the Fig. 5), due to the introduction of attention mechanism, the convergence speed of SMAGRU during model training has been significantly improved compared with GRU without self multi-head attention. In addition, due to that self multi-head attention

enormously enhance the input features, the accuracy during training and validation of SMAGRU is also much higher than that of GRU without attention.

2) FINANCIAL TIME SERIES FEATURES

In this section, we attempt to assess the performance of GRU recurrent networks without bidirectional processing (SMAGRU) and with bidirectional processing GRU (SMABiGRU). The comparison results in Tab. 6 shows that SMAGRU performs better than SMABiGRU in classification accuracy. Especially when tackling sparse, short-term data, SMABiGRU is barely capable of capturing time series sequence features.

In the prior language sequence and image processing tasks, the bi-directional GRU RNNs have harvested excellent prediction achievements. The primary reason lies in the fact that in language or image prediction tasks, it is necessary to rely on the information in the following text to infer the ambiguous expression of the previous text.

However, the above conclusions are all based on the prediction or processing of static language and images. When switching to manage real-time language tasks, such as dialogue tasks, the two-directional mechanism cannot receive due effects or perform operations. The reason is that real-time language processing is a time series problem. In a real dialogue environment, we cannot modify the words we have spoken through reverse reasoning.

The problem of financial prediction or classification based on time series data is the same as that of real-time language processing. The prediction or classification results are obtained based on the cumulative observation of chronological data. Therefore, as the experimental results show, if we follow a bi-directional mechanism in our time series tasks, besides resulting in unsatisfactory classification results, additionally, it will cause unexplainable effects.

D. MODEL GENERALIZATION ABILITY

Even though our architecture performs well on three datasets that consist of different entity values with different labels, which denotes a good generalization ability, considering the constrain of the dataset capacity, moreover, to improve the generalization ability further, we conduct two 10-fold cross-validation experiments in three datasets. In this procedure, the training set and the test set are first mixed and then randomly divided. The model training and testing are performed according to the conventional cross-validation process.

In the first 10-fold of the cross-validation experiment, 90% of the data is taken as the training set each time, and the remaining 10% is used as a test set. The results of the classification experiment are shown in Tab. 7. For comparison, the results of the SMAGRU model in Tab. 5 are listed as a baseline.

From the experimental result in Tab. 7 and Fig. 6, it can be seen that the classification performance of the SMAGRU model in cross-validation is stable; the variance of the test results is small. The accuracy of the D1 set is also close to

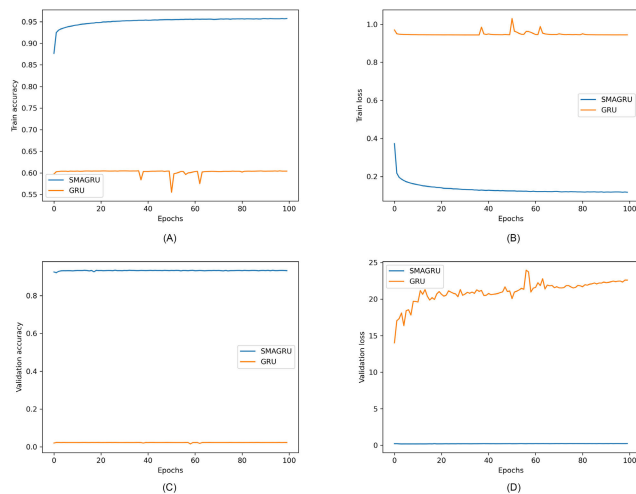
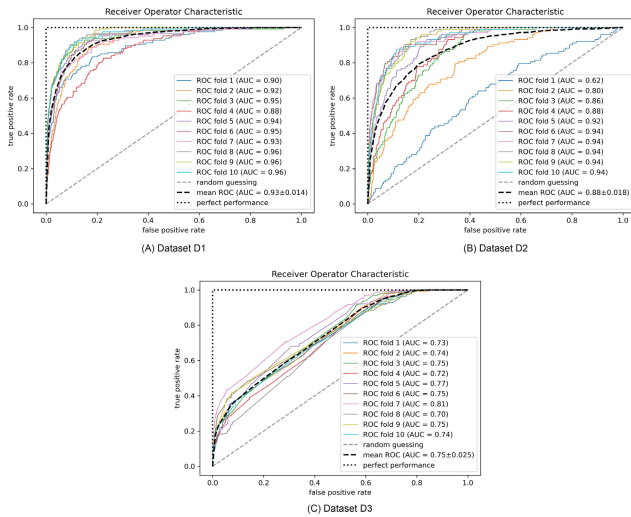


FIGURE 5. Accuracy and Loss curve of SMAGRU and GRU models during training and validation on the D1 dataset.

TABLE 7. Results of two 10-fold cross-validation experiments.

CV	D1	D2	D3
10-fold (90%-10%)	0.93±0.014	0.88±0.018	0.75±0.025
10-fold (10%-90%)	0.70±0.025	0.57±0.031	0.72±0.026
Baseline	0.9333	0.9370	0.9068

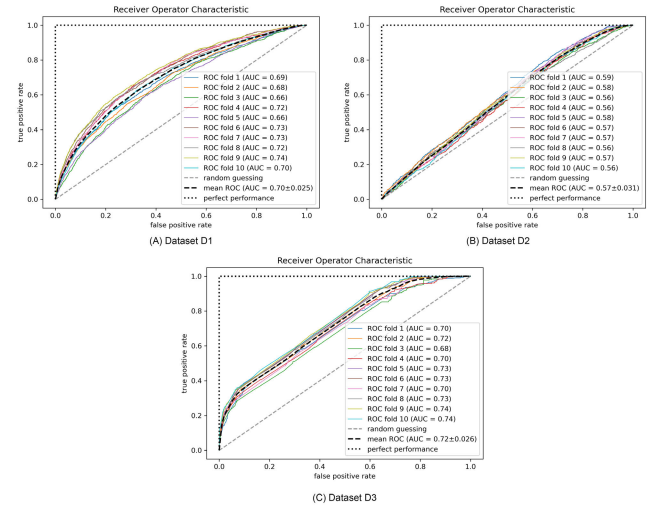
**FIGURE 6.** The ROC curve of 9-1 10-fold cross-validation experiment.

the test results on the original data set. Since the proportion of the training data of the cross-validation experiment is much larger than the proportion of the test data, in order to further evaluate the generalization ability of the SMAGRU model in real application scenarios, we design the following 10-fold validation experiments.

In the second experiment, in each round of verification, only 10% of the data is selected as the training set, and the remaining 90% is used as the test set. The experimental results in Tab. 7 and Fig. 7 show that the accuracy of the SMAGRU model is reduced by about 23%, 31% on D1 and D2, only 3% on D3 compared with the 9-1 cross-validation experiment, indicating that the model can still work effectively in the case of sparse data. Our model contains good generalization ability and robustness. Considering that the experimental conditions are more demanding, according to the above cross-validation experiment results, it can be expected that the SMAGRU model can cope with various data conditions in practical applications.

E. EFFECT OF INPUT FINANCIAL ELEMENT SELECTION

In traditional economics and machine learning methods, most adopt statistical methods to select a cluster of data that is more statistically significant for a rating in financial analysis and credit rating. In this experiment, based on excluding ratio analysis, we also try to select a cluster of financial elements as model input through variance analysis to observe the result comparison between our set of variables and selected set of variables.

**FIGURE 7.** The ROC curve of 1-9 10-fold cross-validation experiment.

For this purpose, one-way ANOVA is utilized to verify whether the differences between different rating classes were significant in each financial element in the D1 set. If the difference is significant ($p < 0.05$), the financial element will be considered significantly impactful concerning the credit rating. We have put data of 88 financial elements to analyze, and the most significant elements are listed in Tab. 8, which are further put into SMAGRU to compare the forecasting accuracy with a cluster of financial elements our model adopts. We conduct the same procedure on SMAGRU with values of ANOVA cluster variables. The comparison of rating classification results is shown in Tab. 9.

TABLE 8. The selected financial statement elements by one-way ANOVA.

Financial statement element	F	Sig.
Earnings Per Share	196.786	0
Net Profit After Deducting	145.178	0
Non-recurring Items	129.421	0
Net Profit Attributable to Parent	111.071	0
Net Profit	109.019	0
Consolidated Total Revenue	106.663	0
Operating Profit	105.556	0
Gross Profit	73.184	0
Total Revenues	68.244	0
Income Tax Expense	61.743	0
Expenses	57.481	0
Cost of Goods Sold	37.304	0
Total Owner's Equity	22.383	0
Salaries and Wages	18.053	0
Service Revenue	15.703	0
Cash Inflow From	13.618	0
Operating Activities	9.792	0
Cash Outflow From	6.986	0
Operating Activities	6.537	0

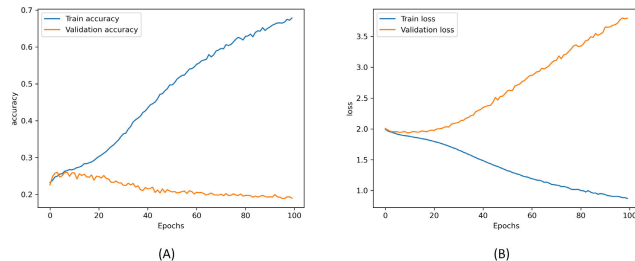
Tab. 9 shows that model with variables selected by ANOVA performs relatively poorly in classification accuracy while

TABLE 9. Comparison of rating classification accuracies from different input variables.

Variable sets	Train	Validation
ANOVA cluster	0.6727	0.1935
Our cluster	0.9568	0.9333

training and obtain substantial low classification results during validation.

When using the variables selected by ANOVA as inputs, although the accuracy during training can reach a satisfactory result, yet it can be seen from (as shown in the Fig. 8) that the accuracy during validation is substantially low and becomes smaller as the iteration increases. That is to say, although statistical methods selecting indicators will promote the improvement of accuracy to a certain extent, in the neural network scenario, it will cause a considerable lack of noise and further significant overfitting.

**FIGURE 8.** Accuracy and Loss curve of SMAGRU with selected element values during training and validation on the D1 dataset.

VI. CONCLUSION

In this paper, we first propose a self-attentive gated recurrent architecture, SMAGRU, in credit rating classification. The model excludes human factors in the evaluation process to the most significant extent benefit from the dynamic weighting module. Each dataset covers as many entities as possible aim to simulate the entire market so that our involved indicators can be more representative. Via extensive experiments, we demonstrate that 1) time series features substantially improve classification performance, 2) self multi-head attention capturing a relevant representation of financial element helps model raise classification accuracy and convergence speed, 3) our model is capable with high-dimension data and large scales of variables. Since the selected variables can cause overfitting, variables covering more financial aspects are more beneficial for rating prediction. Theoretically, all of the financial statement elements can be considered as input of the model, which is not necessary.

However, due to the limitation of paper length, this study focuses on disclosed financial statement data and its corresponding processing architecture instead of factors outside statement sheets. Meanwhile, we noticed that credit agencies analyze corporate credit rating according to its subordinating industry. Our model is capable of the industry, even market

analyses, while due to insufficient data, such analyses did not conduct in our paper. In further research, we will take word embedding into modeling in order to consider the impact of factors outside financial sheets and financial and operating events on credit ratings.

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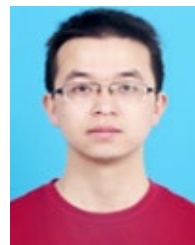
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